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Toward Trust-by-Design in Federated Learning for Healthcare: A Taxonomic Review of Privacy, Explainability, and Accountability

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***Abstract*—Federated learning (FL) has emerged as a leading architecture for privacy-preserving collaborative AI in healthcare, enabling model training across institutions without centralizing sensitive patient data. However, data locality alone does not guarantee clinical trustworthiness. This review critically synthesizes recent literature on trust-enhanced federated learning (TEFL) through a taxonomic framework organized along two orthogonal dimensions: the locus of trust intervention (client, communication, aggregation) and the nature of trust assurance (architectural, cryptographic, interpretive, composite). Within the reviewed studies, the field remains fragmented across four paradigms—architectural privacy, cryptographic guarantees, interpretive transparency, and system accountability—each strengthening one trust dimension while leaving others underdeveloped. Our analysis indicates that explainability is typically validated through visual plausibility rather than clinician-centered utility; privacy mechanisms often provide formal guarantees without deployment-scale evidence; accountability-oriented evidence remains sparse and largely simulation-bound; and within the reviewed corpus, no study was found to demonstrate an integrated trust-by-design architecture that jointly supports privacy, explainability, accountability, and deployment readiness. The central contribution of this paper is to reframe the literature from isolated trust mechanisms to the need for integrated trust-by-design systems for collaborative healthcare AI.**

***Index Terms*—Federated learning; healthcare; trustworthy AI; privacy-preserving; explainable AI; differential privacy; auditability; accountable AI; clinical deployment; trust-by-design**

I. Introduction

AI has advanced from experimental applications to a revolutionary force in healthcare, showing significant influence in clinical decision support, diagnosis, and prognosis [1], [2]. Strong models require large, diverse datasets dispersed across institutions, but healthcare data are very sensitive and subject to ethical and privacy constraints, thus data access is still essentially restricted. Federated learning (FL), which enables cooperative training in which only model updates are shared and the raw data is kept decentralized, is the main architectural option [3], [4]. This offers a privacy-aware foundation for multi-institutional learning [5], especially within cross-silo medical environments [6].

However, conventional FL is insufficient for clinical AI. While it ensures data locality, it lacks the interpretability and accountability required for deployment, as well as built-in defense against adversary manipulation and update-related information leakage. Non-IID data and system heterogeneity are identified as main challenges in conventional FL methods [3], while recent reviews highlight persistent barriers including privacy vulnerabilities, weak validation, and a stark lack of real-world implementation—only 5.2% of reviewed studies report practical application [6], [7].

These limitations motivate the perspective used in this paper, which we call *trust-enhanced FL (TEFL)*. In this view, trust is treated as a design objective rather than as an automatic consequence of decentralization. In healthcare, trust is multi-dimensional, including privacy, explainability, security, robustness, transparency, accountability, and deployability [8], [9]. The literature on explainable AI further clarifies that trust is a calibrated response, not a binary outcome [10].

Recent healthcare FL studies clearly recognize these concerns, but the work is still moving in separate directions. Studies integrate explainability [11], [12], employ formal privacy mechanisms [13], or investigate auditability and update provenance [14], but these developments proceed in parallel, lacking a unified framework. The reviewed literature does not yet provide a coherent understanding of *where* trust mechanisms are inserted—client, communication, or aggregation—and *how* trust is established—architecturally, cryptographically, or interpretively.

To address this gap, we propose a taxonomy built around two dimensions: where the intervention occurs and what kind of trust assurance it provides. The framework yields four paradigms—architectural privacy, cryptographic guarantees, interpretive transparency, and system accountabil-

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ity—revealing that the central challenge is not optimizing individual mechanisms but integrating across paradigms to achieve trust-by-design. More specifically, we: (i) propose a taxonomy that shows the field's conceptual structure; (ii) compare different approaches across loci, assurances, and trade-offs; (iii) find gaps in the literature; and (iv) outline a unified agenda for integrated architectures that provide formal privacy, clinically validated explainability, verifiable accountability, and operational feasibility. The more important question is no longer whether data can stay local, but how distributed model development can be made trustworthy by design.

This paper is structured as follows. The fundamental concepts and background of the field are introduced in Section II. A taxonomy is developed in Section III. Cross-paradigm trade-offs are examined in Section IV. The quality of available evidence and the degree of readiness for deployment are assessed in Section V. Future research directions and key unresolved challenges are discussed in Section VI. The conclusions of this study are summarized in Section VII.

II. Conceptual and Analytical Foundations of Trust in Federated Healthcare AI

A. FL in Healthcare: Architectural Principles and Collaboration Models

In healthcare, FL mainly works as an architectural response to a basic problem: AI models need large amounts of data, while privacy rules severely limit how medical data can be shared. In the usual FL setup, several clients train a shared model under server coordination, but the raw data remain at their original sites [3]. This arrangement suits healthcare because collaboration often takes place in cross-silo settings that involve a limited number of institutions, such as hospitals and research centers.

Figure 1 illustrates this cross-silo arrangement in such a way that each institution keeps its own patient data records, and a central server coordinates the training by collecting model updates rather than raw records. As a result, the design reduces the pressure to centralize sensitive data. However, decentralization alone does not address the broader issue of building clinical trust.

Most of the reviewed studies follow this institution-based pattern. For example, Markkandan et al. [15] presented a framework in which healthcare institutions preprocess data locally, train models on-site, and send only gradients to a coordinating server. Similarly, Agrawal et al. [11] and Latif et al. [16] used FL to build diagnostic models for chest X-rays and brain MRI data without exchanging patient images. A more advanced real-world example appears in [12], where federated training is carried out across three clinical centers rather than being limited to simulation. These studies show that FL has become a structured way to support distributed medical AI across different institutions. The main issue, then, is no longer distributed optimization alone, but how to build trust in a collaborative and heterogeneous environment.

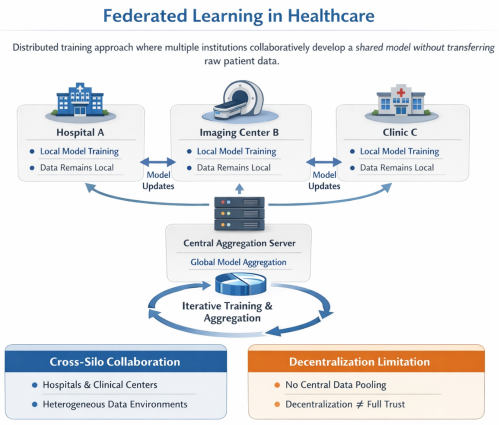


Fig. 1. Conceptual Architecture of Cross-Silo FL in Healthcare Environments

B. The Notion of Trust in Medical Artificial Intelligence

Trust in medical AI is increasingly conceptualized as a multidimensional socio-technical condition rather than a direct technological output. The trustworthiness, in this context, denotes the degree to which a system is considered clinically acceptable, interpretable by its intended users, operationally safe, and compatible with established healthcare workflows. This conceptualization is consistent with the NIST AI Risk Management Framework, which characterizes trustworthy AI as arising from the integration of reliability, safety, security, resilience, accountability, transparency, explainability, and privacy-enhancing capabilities [17]. In parallel, the World Health Organization (WHO) guidance specifies that the design, development, and deployment of clinical AI systems must be grounded in ethical principles, human rights, and the protection of individual autonomy, supported by strong transparency measures and meaningful human oversight [18]. Importantly, the primary objective in establishing trust is not to maximize user confidence in itself, but to promote calibrated trust [10]. By aligning practitioner reliance with the actual capabilities and limitations of the system, calibrated trust supports informed, evidence-based decision-making and mitigates the twin risks of automation bias and unwarranted algorithmic aversion. Consequently, trust can be framed and evaluated as a set of concrete measurable dimensions within federated systems.

C. Core Trust Dimensions in Federated Healthcare Systems

Figure 2 shows that trust in AI in federated healthcare depends on several connected criteria that operate within the larger socio-technical setting. In other words, trust should be understood as a structured condition of clinical acceptability.

Privacy and Confidentiality: Even though privacy is the main reason for adopting FL in this field, it involves more than keeping data locally and requires protection against data and gradient leakage. We found that this ranges from local privacy alone [19] to formal methods such as differential privacy [13] and homomorphic encryption [20].

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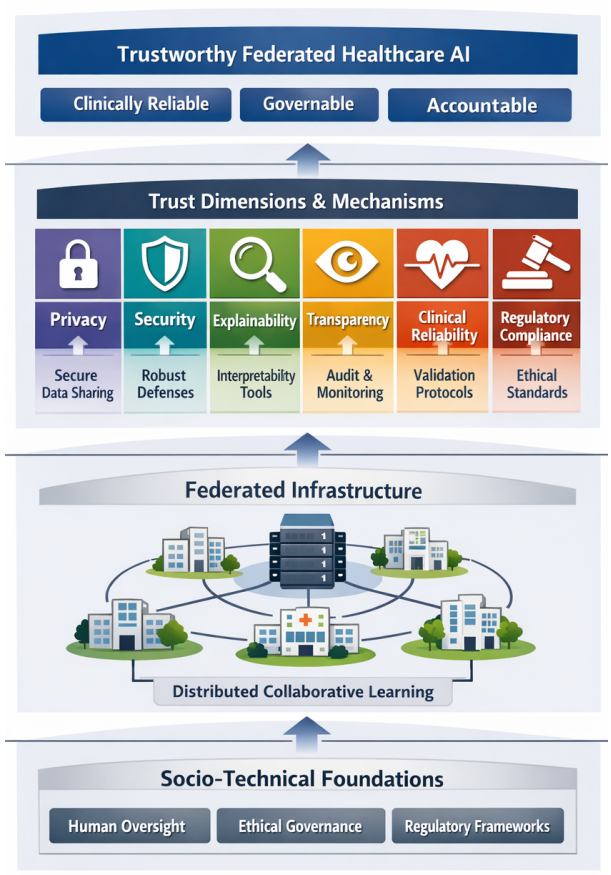


Fig. 2. Socio-Technical Framework for Trustworthy Federated Healthcare Artificial Intelligence

Security and Resilience: The current literature addresses this through robustness testing against label-inference attacks [21] and through blockchain-supported coordination that strengthens integrity monitoring and post-hoc traceability of collaborative updates [14]. Here, *traceability* denotes the ability to reconstruct the history of model updates, data-flow events, or coordination actions across the FL lifecycle.

Explainability and Interpretability: Because FL models are often opaque, they need mechanisms that help clinicians understand how decisions are made. In the literature, they use methods such as Grad-CAM [16], [22], LIME [23], and Integrated Gradients [12] to improve practical understanding rather than visual appeal alone. In our terminology, *explainability* refers specifically to methods that make model predictions or learned representations understandable to users, for example through heatmaps, feature attribution, or biologically grounded evidence.

Transparency, Traceability, and Auditability: This principle concerns the degree to which a system's documentation, developmental logic, and validation procedures are accessible for rigorous evaluation. Our review indicates that this requirement is most effectively addressed through blockchain-assisted architectures. Such designs leverage decentralized ledgers to maintain immutable logs of update histories, coordination events, and collaborative model management processes, thereby providing a verifiable audit trail for the entire system lifecycle [14]. In this framework, *transparency* denotes the openness of the system's design logic, documentation, assumptions, and validation procedures; *traceability* signifies the ability to reconstruct the history of model updates, data-flow events, or coordination actions; and *auditability* refers to the capacity to inspect and verify these records after training or deployment.

Clinical Dependability and Human Oversight: Trust is contingent upon the extent to which a system supports secure and reliable clinical reasoning while concurrently maintaining substantive and effective human oversight. This dimension is consistent with the prevailing bioethical and professional frameworks, which maintain that AI should act as a collaborative instrument designed to enhance, rather than replace, clinical judgment and medical expertise. Moreover, it underscores that true reliability can be demonstrated only through a methodologically rigorous assessment carried out in the actual healthcare settings where the technology is intended to be used [24], [25]. This aspect relates to *accountability*, which concerns the assignment of responsibility for system behavior, failures, update provenance, or governance decisions.

Regulatory Compliance and Governance: The realization of trust is fundamentally contingent upon the existence of a robust framework of legal and regulatory compliance. The reviewed literature consistently underscores this association, demonstrating that the implementation of transparent governance mechanisms, coupled with strict compliance with established data protection and privacy regimes—such as HIPAA and the GDPR—is indispensable. These elements constitute the necessary conditions for the responsible integration of AI systems in healthcare settings [13], [25]. In our conceptual framework, *governance* refers to the institutional, legal, ethical, and regulatory structures that guide the responsible development and use of the system.

These six dimensions provide a rigorous framework for analyzing and evaluating TEFL systems. They show that trust is not an automatic result of decentralization but an emergent property that must be deliberately designed and built into the system's architecture.

D. A Conceptual Model of Trust-Enhanced Federated Healthcare AI

Figure 3 shows TEFL as a layered (multi-tier) architecture, designed to accommodate data heterogeneity and broader systemic constraints and challenges. The intermediate layers provide core trust mechanisms (privacy, security, robustness) as a protective boundary, while explainability, interpretability, transparency, and auditability keep the system intelligible, inspectable, and traceable. The upper layers—clinical integration, human oversight, governance, and regulatory standards—form the socio-technical apex, aligning with frameworks that prioritize accountability and oversight [24], [25]. Collectively, this architectural model shows that trust is not

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fixed or pre-packaged; it is a dynamic, emergent property built progressively from lower architectural layers upward. A system may preserve privacy and perform well technically yet still fail to earn clinical trust if it does not fit existing workflows, support clinician oversight, or meet governance expectations. In this sense, TEFL should be treated as a multi-layer design paradigm in which decentralized learning becomes clinically credible only when technical design, privacy and security measures, interpretability tools, auditability support, human oversight, and regulatory governance are developed together.

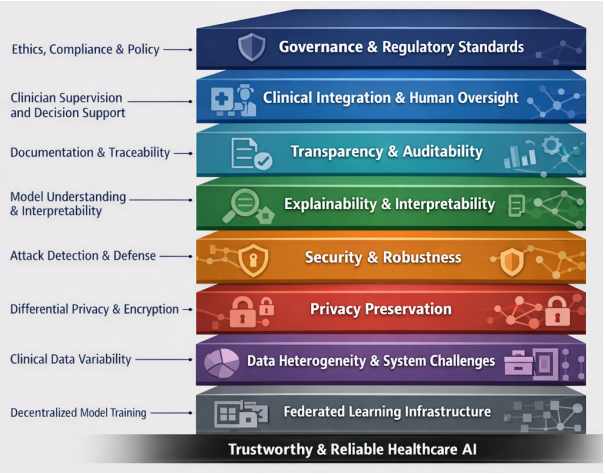


Fig. 3. Layered Framework for Trustworthy and Reliable Federated Healthcare Artificial Intelligence

III. A Two-Dimensional Taxonomy of Trust-Enhanced FL in Healthcare

This paper develops its taxonomy through analytical synthesis of representative recent studies on privacy, explainability, and accountability in FL for healthcare. A coherent taxonomy of trust-enhanced FL must do more than catalogue existing approaches; it must reveal the conceptual structure of the field, illuminate relationships among methods, and expose the assumptions that shape research trajectories. We propose a taxonomy organized along two orthogonal dimensions: the *locus of trust intervention*—where in the FL pipeline trust mechanisms are inserted—and the *nature of trust assurance*—whether trust is achieved through architectural constraints, cryptographic guarantees, or interpretive transparency. This framework, synthesized in Table I, yields four principal paradigms that structure the current literature and, more importantly, reveal its foundational limitations. The two dimensions were derived a priori from existing trust frameworks and refined through inductive mapping of the literature.

In practice, trust mechanisms enter the FL pipeline at three main points. The *client locus* addresses local data and model updates before aggregation; it is the most densely populated region, encompassing both foundational architectural privacy and the majority of explainability mechanisms. The *communication locus* governs transmission between clients and server, uniquely positioned to address passive eavesdropping and interception threats through cryptographic mechanisms. The aggregation stage is where issues such as poisoning, free-riding, and coordination integrity become especially important, yet it is still less studied than the client side. Orthogonal to locus is the nature of trust assurance. *Architectural assurance* relies on design choices that rule out certain failures by construction. In other words, the claim to trust is based on what the system prevents from happening. *Cryptographic assurance* provides trust through mathematical provability, shifting reliance from design assumptions to theorems. *Interpretive assurance* produces trust through transparency and human comprehensibility, enabling clinical users to assess model behavior. *Composite assurance* combines several kinds of protection in one system, but the way these mechanisms interact is still not well understood.

The intersection of these dimensions yields four principal paradigms. *Architectural privacy by decentralization* occupies the client locus with architectural assurance. Studies within this paradigm [19], [26], [27] rely on data locality as the primary trust mechanism, claiming that exposure by centralization is precluded by design. The paradigm's strength lies in its minimal overhead and conceptual simplicity, but it provides no protection against inference attacks on shared updates, and its trust assumptions remain implicit and unverifiable.

Cryptographic privacy guarantees occupy the communication and aggregation loci with cryptographic assurance. Communication-channel encryption [15], [28] provides *ex ante* assurance that intercepted updates reveal no information, though it assumes trusted key management infrastructure. Aggregation-level differential privacy [13], [21], [29] provides quantifiable privacy budgets against inference attacks, introducing a fundamental privacy-utility trade-off while assuming a trusted aggregator. Hybrid cryptographic approaches [20] combine multiple mechanisms across loci for composite assurance, incurring the highest computational complexity while leaving mechanism interactions undercharacterized.

Interpretive transparency occupies the client locus with interpretive assurance. Visual plausibility mechanisms [11], [16], [22], [23], [30] generate heatmaps or superpixel highlights that align with clinical expectations, but conflate plausibility with correctness—a heatmap highlighting the lung region guarantees neither sound reasoning nor clinical utility. Feature-attribution approaches [31] decompose decisions into contributory weights, yet attribution does not imply causation. Biologically grounded explanations [12] link predictions to quantifiable biological features, representing the most rigorous interpretive paradigm, though validation remains analytical rather than clinical, with no clinician-in-the-loop assessment.

System-level accountability is the least common paradigm in the reviewed corpus and is mainly realized through blockchain-assisted auditability and traceability. Accountability-oriented designs [14] focus on post-hoc inspection of update flows, coordination events, and system behavior in privacy-sensitive multi-institutional settings. They contribute more to traceability and integrity than to preventive protection, and current evidence is narrow, simulation-based,

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TABLE I
A TAXONOMIC FRAMEWORK FOR TRUST-ENHANCED FL IN HEALTHCARE

Paradigm	Locus of Intervention	Nature of Trust Assurance	Representative Studies	Core Trust Claim	Fundamental Limitation
Foundational Paradigm: Architectural Privacy					
Architectural Privacy	Client	*Ex negativo* (trust by constraint)	[19], [26], [27]	Data never centralized; exposure precluded by system design	No formal guarantees against inference attacks; trust assumptions implicit and unverifiable
Cryptographic Paradigm: Mathematical Guarantees					
Communication-Channel Encryption	Communication	*Ex ante* (trust by secrecy)	[15], [28]	Mathematical infeasibility of information extraction from intercepted updates	Substantial computational overhead; assumes trusted key infrastructure
Aggregation-Level Differential Privacy	Aggregation	*Ex post* (trust by proof)	[13], [21], [29]	Quantifiable (ϵ, δ) bounds against inference attacks; formal privacy accounting	Privacy-utility trade-off; assumes trusted aggregator; narrow attack coverage
Hybrid Cryptographic	Multi-locus	*Composite* (trust by layering)	[20]	Combined guarantees across communication and aggregation layers	Highest cryptographic complexity; additive overhead; integration not formally characterized
Interpretive Paradigm: Transparency-as-Trust					
Visual Plausibility	Client	*Phenomenological* (trust by resemblance)	[11], [16], [22], [23], [30]	Explanations visually align with clinical expectations; attention maps highlight plausible regions	Plausibility conflated with correctness; no causal fidelity guarantees; clinician validation absent
Feature-Attribution	Client	*Reductionist* (trust by decomposition)	[31]	Model decisions decomposed into contributory feature weights; clinical predictors identified	Attribution does not imply causation; no validation of explanation utility in clinical tasks
Biologically Grounded	Client	*Evidentiary* (trust by mechanism)	[12]	Explanations link to quantifiable biological features; cross-center feature stability demonstrated	Analytical validation only; no clinician-in-the-loop assessment; mechanistic claims unverified
Accountability Paradigm: Verifiable Collaboration					
Update Traceability and Auditability	Aggregation	*Forensic* (trust by audit)	[14]	Blockchain-assisted coordination supports post-hoc traceability of update flow, system events, and collaborative model management	Sparse evidence base; simulation-bound; limited integration with formal privacy and interpretability
Unoccupied Paradigm: Integrated Trust-by-Design					
Theoretically Possible but Empirically Vacant	*Multi-locus*	*Emergent* (trust by co-design)	None	Simultaneous formal privacy guarantees, clinically validated explainability, system accountability, and deployment readiness	Within the reviewed studies, no paper was found to integrate all dimensions; represents the central research gap

and weakly integrated with formal privacy or interpretability.

Table I synthesizes the four paradigms by main intervention locus and primary assurance type. The categories are analytical, not strictly exclusive, and some systems span multiple loci or trust mechanisms. The taxonomy highlights that the client locus is crowded, while the aggregation locus, despite its central role in coordination integrity, remains underexplored. Architectural, cryptographic, interpretive, and composite assurances represent fundamentally different epistemological commitments that are not interchangeable. Interpretive paradigms operate largely within phenomenological plausibility, with clinically validated evidentiary approaches virtually absent. Accountability-oriented evidence is especially sparse, with the reviewed corpus containing auditability and provenance mechanisms but no convincing multi-locus trust infrastructure that integrates accountability with privacy, interpretability, and deployment readiness.

Most significantly, the taxonomy reveals what does not exist: a paradigm occupying the multi-locus space with composite assurance that integrates formal privacy guarantees, clinically validated explainability, system accountability, and deployment readiness. Within the reviewed studies, no paper was found to simultaneously demonstrate formal privacy guarantees, clinically validated explainability, verifiable accountability, and deployment readiness in a single integrated architecture. The field operates within distinct silos—architectural privacy, cryptographic guarantees, interpretive transparency, or system accountability—without architectures that co-design these dimensions from the outset. This is the central gap in the literature. Future systems need to treat privacy, explainability, security, and accountability as linked design requirements rather than separate add-ons. The taxonomic framework thus serves not merely to classify existing work but to reveal the conceptual architecture of the field, expose its hidden assumptions, and illuminate the path toward genuinely trustworthy federated healthcare AI.

IV. COMPARATIVE TRADE-OFFS ACROSS TRUST PARADIGMS

Trust in trust-enhanced FL arises not from optimizing mechanisms in isolation but from managing tensions across four paradigms: architectural privacy, cryptographic guarantees, interpretive transparency, and system accountability. Each defines trust differently and entails distinct trade-offs

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TABLE II
EXPLAINABILITY STRATEGIES AND THEIR PERFORMANCE IMPLICATIONS IN FL-BASED CLINICAL MODELS

Study	Explainability Strategy	Representative Techniques	Performance Gain / Retention	Integration Level in FL	Effect on Model Performance
[22]	Post-hoc heatmap-based visual explanation	Grad-CAM	Competitive federated performance retained (86.5–87.1% accuracy)	Low to moderate	Performance retention alongside improved visual interpretability
[23]	Local surrogate and superpixel-based explanation	LIME	Strong federated performance retained (92.08% on ISBI2016, 94% on ISBI2017)	Moderate	Preserved model utility with enhanced local interpretability
[16]	Hybrid heatmap- and saliency-based visual explanation	Grad-CAM + Saliency maps	Good federated performance retained (approx. 94% accuracy)	Moderate	Preserved predictive performance and improved qualitative interpretability
[11]	Post-hoc heatmap-based visual explanation	Grad-CAM	Acceptable performance retained (88% accuracy; 89.66% F1-score)	Low to moderate	Enhanced transparency with retained predictive utility
[12]	Biologically grounded quantitative explanation	Integrated Gradients + quantitative cellular feature analysis	Cross-center predictive performance with analytically grounded biological explanation (overall AUROC 0.82)	High	Explainability linked to analytical validation and cross-center robustness
[30]	Multi-method local visual explanation pipeline	LIME + Grad-CAM	Improved pneumonia-prediction performance (97.61% accuracy, 98.36% F1-score)	Moderate to high	Strong overall performance, but independent contribution of each XAI component not quantified
[31]	Feature-attribution explainability for structured clinical FL	SHAP	Strong performance retained for CAD prediction (83.21% with privacy, 84.49% without)	High	Improved feature-level transparency with modest performance decline under privacy-preserving settings

in interpretability, privacy strength, model utility, cost, and deployability. This section systematically analyzes these trade-offs, showing why no single paradigm ensures comprehensive clinical trustworthiness and why integrating paradigms remains the key unresolved challenge.

A. Trade-Offs Within the Interpretive Paradigm: Transparency Versus Validation Rigor

The interpretive paradigm faces a core tension between explanatory richness and rigorous validation. As Table II shows, visual methods like Grad-CAM offer efficient transparency and preserve high predictive accuracy in federated pipelines [11], [22]. Yet their interpretability rests largely on qualitative visual inspection rather than systematic evaluation. This highlights a key trade-off: visual explanations are accessible, but plausibility alone does not ensure explanatory fidelity or clinical utility.

As integration within the interpretive paradigm deepens, studies such as [12] and [31] show the potential for clinically meaningful insights using biologically grounded feature analysis and SHAP-based attribution. Yet this explanatory richness brings substantial methodological complexity and underscores ongoing validation challenges. For instance, [12] reports a strong AUROC of 0.82 across three clinical centers with quantitative feature validation but lacks clinician-in-the-loop evaluation. Likewise, [31] provides feature-level transparency for coronary artery disease prediction but omits real-world workflow validation.

The main trade-off is not between accuracy and interpretability—accuracy remains high—but between how deep explanations go and how feasibly they can be rigorously validated. The interpretive paradigm still lacks standardized methods to show that more sophisticated explanations actually improve clinical judgment at the bedside.

B. Trade-Offs Within the Cryptographic Paradigm: Privacy Strength Versus Operational Burden

The cryptographic paradigm, applied at communication and aggregation points with *ex ante* or *ex post* assurance, highlights a core trade-off between the formal strength of privacy guarantees and the operational effort needed to implement them. Table III summarizes the privacy strategies and trade-offs in the reviewed studies—covering privacy protection, attack robustness, scalability, and deployment feasibility. It shows that all approaches maintain high utility (accuracies above 95%), though their privacy protections differ substantially.

At the lowest end of the assurance spectrum, architectural privacy methods [19], [26], [27] achieve optimal efficiency and scalability but provide only implicit, unverifiable protection against inference attacks. Their trust claim is negative—data is not centralized—rather than positive assurance. At the communication locus, homomorphic encryption [15], [28] provides strong *ex ante* guarantees against interception but introduces substantial computational overhead and assumes trusted key management infrastructure. Differential privacy at the aggregation locus [13], [21], [29] offers formal, quantifiable privacy budgets with provable bounds against inference attacks, yet this assurance comes at the cost of a privacy-utility trade-off and the assumption of a trusted aggregator. Hybrid cryptographic approaches [20] layer multiple mechanisms for composite assurance, achieving the strongest formal guarantees but incurring the highest complexity and leaving mechanism interactions undercharacterized. The key insight from Table III points to a clear pattern: stronger cryptographic protection usually comes with greater operational burden. Methods with formal guarantees (DP, HE) remain simulation-bound with limited scalability evidence; methods with demonstrated efficiency (locality-based) lack formal protection. No approach yet demonstrates both strong formal privacy guarantees and

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TABLE III
STRUCTURED PRIVACY STRATEGIES AND MULTI-DIMENSIONAL TRADE-OFFS IN FEDERATED CLINICAL LEARNING

Study	Privacy strategy	Representative mechanisms	Perf. gain / retention	Privacy protection	Scalability	Attack robustness	Deployment feasibility
[19]	Privacy by decentralization	FL data locality, FedDEO + FedAvg	96.98% acc.; improved vs. baseline	Implicit / low-formality; no DP/HE	Moderate; simulated multi-client	Not explicitly evaluated	Limited; no real institutional evidence
[28]	Encrypted model-update protection	Homomorphic Encryption, FedAvg + IKMC	98.6% acc.	High at update/communication level; no formal budget	Limited to moderate; encryption adds cost	Partially addressed; leakage risk mitigated	Limited; simulated hospital setting
[29]	DP with secure aggregation	DP, Secure Aggregation, CNN-LSTM FL	Near-centralized utility retained (∼1-2% decline)	High; explicit DP-based protection	Moderate; acceptable efficiency as devices increased	Partially addressed; model inversion targeted	Limited; fully simulated environment
[26]	Lightweight locality-based	Local data retention, weighted FedAvg	95.70% (3 clients) to 96.56% (5 clients)	Implicit / low-formality	Moderate to high; lightweight design	Not explicitly evaluated	Limited; simulated multi-center setting
[15]	Encrypted aggregation privacy	PHC, encrypted parameter aggregation	∼10% avg. acc. improvement; 30% overhead reduction	High at aggregation level; no formal accounting	Moderate; PHC computation 150 ms	Conceptually addressed; leakage/misuse discussed	Limited; simulated collaboration
[27]	Decentralized privacy-aware	Data locality, adaptive FedAvg/FedSGD	Improved/stabilized learning under heterogeneity	Implicit to moderate; privacy via decentralization	Moderate; adaptive aggregation improves convergence	Conceptually addressed only; no formal testing	Limited; experimental framing
[21]	DP-based privacy hardening in SplitFed	DP noise, SplitFed architecture	Utility retained; primary gain was privacy hardening	Moderate to high; strong against label inference	Limited; time/space complexity underdeveloped	Strong for label inference specifically	Low to limited; preliminary interpretation
[13]	Formal DP-based	Gradient clipping, Gaussian noise, FL-DP	96.1% acc.; slight privacy-induced degradation	High (formal); $\epsilon = 1.9, \delta = 10^{-5}$	Limited to moderate; acknowledged overhead	Moderate to strong; membership inference addressed	Limited; experimental, no hospital-scale
[20]	Hybrid multi-layer	DP + HE, encrypted exchange, adaptive weighting	12% improvement over centralized baseline	Very high; multi-layer protection	Moderate; encryption adds nontrivial burden	Partially addressed; stronger protection	Moderate; among more translational, but complex

proven scalability in real-world multi-institutional healthcare settings.

C. Trade-Offs Across Paradigms: Fragmentation as the Central Barrier

The field is not simply split between privacy and explainability. It is more deeply fragmented across four paradigms that make different assumptions about where trust should be built and how it should be justified. Table IV synthesizes these cross-paradigm trade-offs, exposing why no existing approach achieves comprehensive clinical trustworthiness. The readiness labels in Table IV are comparative analytical judgments and are interpreted according to explicit analytical criteria. "Limited" refers to evidence based mainly on single-dataset testing, internal validation, qualitative explanation, or narrow simulation. "Moderate" indicates the presence of additional validation elements, such as external testing, partial cross-site evaluation, formal privacy analysis, or targeted security testing, but without clinician-in-the-loop assessment or operational deployment. "Strong" refers to evidence that combines robust internal validation with external or cross-site evaluation and a clearer link between technical performance and clinical relevance. "Simulation-bound" denotes studies whose evaluation remains restricted to simulated clients, public datasets, or experimental settings without operational healthcare deployment. "Clinical readiness" is used as a comparative analytical label rather than a regulatory determination; it reflects the extent to which a study reports multi-site validation, privacy and security testing, explanation validation, clinician involvement, workflow compatibility, and deployment feasibility.

Architectural privacy is optimized for simplicity and efficiency, but sacrifices formal assurance. Cryptographic guarantees provide mathematical certainty at the cost of operational burden and infrastructure dependencies. Clinical plausibility is attained through interpretive transparency, but it lacks validation frameworks that would turn plausibility into proof. Although system accountability makes collaboration verifiable, it adds complexity that hasn't been validated in practical situations. Each paradigm excels along one dimension of trust while leaving others critically underaddressed.

This fragmentation helps explain why so many studies solve one trust problem while leaving the others largely unresolved. Studies pursuing interpretive transparency [11], [16], [22], [23], [30], [31] achieve high performance and clinical plausibility but provide no formal privacy guarantees beyond baseline data locality. Studies pursuing cryptographic guarantees [13], [20], [28], [29] achieve strong privacy protection but lack clinically validated explainability and remain simulation-bound.Accountability-oriented evidence, represented in our corpus primarily by blockchain-assisted traceability and auditability mechanisms [14], demonstrates post-hoc inspection and coordination support but not integrated formal privacy or validated interpretability.

D. The Unresolved Problem: Integration Costs and Emergent Trade-Offs

The lack of integrated systems is not just a gap in coverage. It also reflects unresolved questions about what happens when several trust mechanisms are combined in one system. Cryptographic privacy mechanisms may degrade the fidelity of explanations—DP noise added to protect privacy could obscure the

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TABLE IV
CROSS-PARADIGM TRADE-OFFS IN TRUST-ENHANCED FL

Paradigm	Primary Trust Claim	Characteristic Strength	Characteristic Weakness	Operational Cost	Clinical Readiness
Architectural Privacy	Data never centralized	Minimal overhead; simple implementation	No formal guarantees; inference attack vulnerability	Low	Limited
Cryptographic Guarantees	Provable privacy bounds	Quantifiable, verifiable protection	High computational burden; trusted infrastructure assumptions	High	Simulation-bound
Interpretive Transparency	Decisions are understandable	Clinical plausibility; transparency	Validation absent; plausibility conflated with correctness	Low to moderate	Prototypical
System Accountability	Collaboration is traceable and auditable	Post-hoc verification; provenance logging	Sparse evidence base; limited preventive protection; simulation-bound	Moderate to high	Experimental
The Unoccupied Paradigm: Integrated Trust-by-Design					
Integrated Systems	Trust emerges from co-design	Simultaneous privacy, transparency, accountability, deployability	No existing implementations; interaction effects unknown	Unknown	None

very patterns that explainability techniques aim to visualize. System accountability mechanisms that create immutable audit trails may conflict with privacy requirements that limit data retention. Interpretive transparency that reveals model decision boundaries may increase vulnerability to inference attacks. These potential interaction effects remain unexplored precisely because the field has not yet produced systems that integrate across paradigms.

The next step for the field is therefore not small improvements inside one paradigm, but architectures that can manage trade-offs across paradigms. Such architectures must determine, for example, how much privacy can be sacrificed for explanation fidelity, or how much auditability can be compromised for computational efficiency, within clinically acceptable bounds. This requires not only technical innovation but also the development of evaluation frameworks that can assess trade-offs across trust dimensions simultaneously—frameworks that currently do not exist.

The taxonomic framework reveals that trust in federated healthcare AI cannot be achieved by selecting a single paradigm and optimizing within it. Each paradigm's strengths are inextricably linked to its characteristic weaknesses. Progress toward clinically deployable systems therefore requires moving beyond paradigm-bound optimization to the co-design of multi-locus, composite-assurance architectures that can manage the inevitable tensions among privacy, transparency, accountability, and practicality. This is the fundamental challenge that the field must now confront.

V. EVIDENCE QUALITY, CLINICAL TRANSLATION, AND DEPLOYMENT READINESS

The credibility of TEFL is strongly dependent on the methodological rigor with which the four paradigms are assessed. Table V highlights a consistent imbalance: each paradigm primarily reinforces the evaluative criteria it already privileges, rather than addressing the broader set of requirements that are critical for clinical deployment. Architectural privacy investigations typically achieve a moderate level of evidential support via external validation, but rarely incorporate formal privacy analyzes. Cryptographic studies offer rigorous formal guarantees, but their evaluations are limited to simulated settings, with no reported engagement of clinical end users. Interpretive studies show the greatest heterogeneity, ranging from minimal evidence (e.g., purely qualitative visual inspection) to robust support (e.g., quantitative multicenter validation in [12]); however, none of these studies implements clinician-in-the-loop evaluation. Evidence aligned with accountability is particularly limited and consists mainly of isolated examinations of provenance and auditability, constraining the possibility of conducting rigorous comparative syntheses at this evidential level. The overarching pattern is that each paradigm validates its own preferred metrics—such as formal guarantees, predictive performance, or perceived visual plausibility—yet none provides validation spanning the multidimensional criteria required for safe and effective clinical implementation.

A. Validation Gaps Across Paradigms

Table VI makes clear that explainability validation is still methodologically limited. Visual plausibility studies—the majority of the interpretive corpus—validate only that heatmaps highlight expected anatomical regions, conflating plausibility with correctness. Quantitative fidelity metrics, clinician-in-the-loop assessment, and cross-site consistency evaluation are universally absent. Even [12], the strongest example with quantitative feature analysis across centers, lacks clinician reader studies. The interpretive paradigm has no standardized frameworks for assessing whether explanations actually enable sound clinical judgment.

Privacy validation exhibits the inverse asymmetry. Cryptographic studies excel at formal technical validation—DP studies report explicit budgets, encryption studies detail computational overhead—but lack translational evidence. Rigorous DP guarantees are provided by [13], though hospital-scale validation is not performed. A comprehensive privacy architecture is achieved by [20], yet the work remains simulation-bound. Furthermore, standardized benchmarks for evaluating privacy methods across multiple attack vectors and deployment contexts are currently lacking in the field.

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TABLE V
EVIDENCE STRENGTH ACROSS TAXONOMIC PARADIGMS

Study	Paradigm	Internal	External	Cross-Site	Clinician	Formal Privacy	Security Testing	Evidence Strength
[22]	Interpretive (Visual)	Yes	No	No	No	No	No	Limited
[19]	Architectural	Yes	Yes	No	No	No	No	Moderate
[28]	Cryptographic (Encryption)	Yes	Yes	Partial	No	No	No	Moderate
[12]	Interpretive (Biologically Grounded)	Yes	Yes	Yes (real)	No	No	No	Strong
[13]	Cryptographic (DP)	Yes	No	No	No	Yes	Partial	Moderate
[20]	Cryptographic (Hybrid)	Yes	Yes	Partial	No	Yes	Partial	Moderate
[21]	Cryptographic (DP)	Yes	No	No	No	Partial	Yes	Moderate
[15]	Cryptographic (Encryption)	Yes	No	Partial	No	No	No	Limited
[26]	Architectural	Yes	No	Partial	No	No	No	Moderate
[23]	Interpretive (Visual)	Yes	Yes	No	No	No	No	Moderate
[31]	Interpretive (Attribution)	Yes	No	No	No	Partial	No	Moderate

TABLE VI
EXPLAINABILITY VALIDATION ACROSS INTERPRETIVE PARADIGM

Study	Validation Method	Main Limitation
[22]	Qualitative visual inspection	No quantitative evaluation; no external validation
[23]	Qualitative LIME inspection	No clinician validation; no uncertainty modeling
[16]	Qualitative visual only	No external validation; no quantitative metrics
[11]	Visual verification of regions	No clinician validation; no stability analysis
[12]	Quantitative feature analysis	No clinician reader study; no uncertainty modeling
[31]	SHAP attribution	No clinician validation; simulated setting only

B. Translational Barriers

Three translation barriers appear repeatedly across the reviewed studies. *Simulation dependence*: the overwhelming majority of studies are validated in simulated environments with idealized conditions. A rare exception is found in [12], where a real multi-center evaluation is conducted. *Clinician absence*: no study includes clinician-in-the-loop validation, treating trust as a technical property rather than a socio-technical relationship. *Paradigm siloing*: studies validate within their paradigm's preferred metrics—plausibility, formal bounds, predictive performance—without integrating across dimensions. No study demonstrates whether formally private models remain interpretable or whether interpretable models maintain privacy under deployment conditions.

C. Toward Integrated Validation

Real progress will require integrated evaluation frameworks that assess privacy, explainability, accountability, and operational feasibility together rather than separately. Such frameworks must standardize: (i) privacy evaluation across multiple attack vectors with formal guarantees and cost reporting; (ii) explainability validation through quantitative fidelity metrics, clinician task-based assessment, and cross-site consistency protocols; (iii) accountability assessment through audit trail integrity and coordination reliability; and (iv) deployment readiness through real-world pilots across heterogeneous institutions. The field's fragmentation is reinforced by validation frameworks that silo; methodological reform is as essential as technical innovation. Minimum deployment-ready evaluation requires: multi-site validation, privacy attack testing, explanation fidelity metrics, clinician usability studies, and operational benchmarking, none of which are consistently present in the reviewed corpus.

VI. CRITICAL GAPS AND FUTURE RESEARCH DIRECTIONS

The reviewed studies point to one central problem: privacy, explainability, security, and accountability are still handled mostly as separate concerns instead of interconnected design requirements. Addressing this gap requires a more integrated trust-by-design agenda.

a) Integrated Trust Architectures: No reviewed system fully combines multiple loci with composite assurance. Future work should examine how trust mechanisms interact, including trade-offs between privacy, interpretability, and auditability.

b) Clinically Grounded Explainability: Most explainability validation still relies on visual plausibility rather than clinical usefulness. Future research should emphasize quantitative fidelity measures, clinician-centered assessment, and cross-site consistency.

c) Deployable Privacy Frameworks: Formal privacy mechanisms are increasingly available, but their deployment remains limited. Priorities include scalable architectures, stronger benchmarking, and validation in realistic multi-center settings.

d) Accountable Collaboration Infrastructure: Accountability-oriented approaches show the feasibility of provenance logging and post-hoc verification, but evidence remains sparse. Further work is needed on integrated auditability, efficiency, and regulatory alignment.

e) Trust-by-Design Deployment: Trust should be embedded from the outset rather than added later. This requires clinician-centered design, interoperability with hospital infrastructure, and evaluation in practical multi-institutional environments. Overall, progress depends on FL systems that jointly support formal privacy, clinically meaningful explainability, verifiable accountability, and operational feasibility.

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VII. Conclusion

This paper examined FL in healthcare as a trust-engineering paradigm rather than only a distributed alternative to centralized training. Through a taxonomy based on locus of intervention and nature of trust assurance, the reviewed studies were organized into four paradigms: architectural privacy, cryptographic guarantees, interpretive transparency, and system accountability. The analysis shows that existing approaches strengthen particular trust dimensions, but none achieves comprehensive clinical trustworthiness. The main contribution of this paper is to shift attention from isolated trust mechanisms to integrated trust-by-design systems. In the reviewed studies, external validation, cross-site evaluation, clinician participation, and deploymentfocused evidence remain limited, and no single system was found to combine formal privacy guarantees, clinically validated explainability, verifiable accountability, and deployment readiness. Future progress depends on integrated architectures, stronger clinical validation, scalable privacy and accountability mechanisms, and realistic multi-institutional evaluation.

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